UNIVERSITY OF CAMBRIDGE

AI Applied to Finance: Guest Lecture 1

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Heuristic finance

We focus on modern empirical asset pricing research. In particular we are interested to study the dynamic of the risk premium. What is the risk premium? We can define the risk premium as the conditional expectation of the future realised excess return.

$$E(r_{t+1} - r_{t+1,rf}|\mathcal{F}_t) \tag{1}$$

where $r_{t+1,rf}$ is the risk-free rate. Under efficient market hypothesis the risk-premium is zero since the market reflects all available information (no free-lunch). However, empirical studies show that a risk-premium exists and it is time varying.

Denoting by $r_{t+1} - r_{t+1,rf}$ the realised excess return and parametrising $E(r_{t+1} - r_{t+1,rf}|\mathcal{F}_t) = F(x)$ as a function of the predictor variables then the focus of our study can be written as:

$$r_{t+1} - r_{t+1,rf} = F(X) + \epsilon_i \tag{2}$$

with ϵ_i a white noise.

Statistical Decision Theory

Basic theoretical framework to the models we will consider. Mathematically should be formulated more precisely and what we do here is mainly formal. Let $X \in \mathbb{R}^p$ and $Y \in \mathbb{R}$ a real-value random input and output respectively with joint probability distribution $p(X, Y)$. We need to find a function $f : X \to Y$ for predicting Y given the value of X. How do we do it?
We define a loss function $L(Y, f(X))$, for example:

- the familiar residual sum of square (RSS)
  $L_2 = L(Y, f(X)) = (Y - f(X))^2$
- $L_1 = L(Y, f(X)) = |Y - f(X))|$

Statistical Decision Theory

The expected prediction error is defined as:

$$EPE = E(Y - f(X))^2 = \int (Y - f(X))^2 p(dx, dy) \tag{3}$$

where we used the L_2 loss function. Minimising the expected prediction error point wise we get:

$$f(x) = E(Y|X = x) \tag{4}$$

The above equation (conditional mean) is also know as the **regression function** and it is the subject of our study.

Let $\hat{f}(X)$ be a prediction model estimated from a training set τ then the test error is defined as:

$$Err_\tau = E[L(Y, \hat{f}(X))|\tau] \tag{5}$$

and the expected prediction error become:

$$Err = E[Err_\tau] \tag{6}$$

i.e. expectation over 'every possible' τ. While Err_τ is easy to measure since for a given training set we can define:

$$\overline{Err} = \frac{1}{n}\sum_{i=1}^{n} L(y_i, \hat{f}(x_i)) \tag{7}$$

Err is not. Usually you split the data in

- Training set to fit the models (used to measure the test error)
- validation set to estimate prediction error

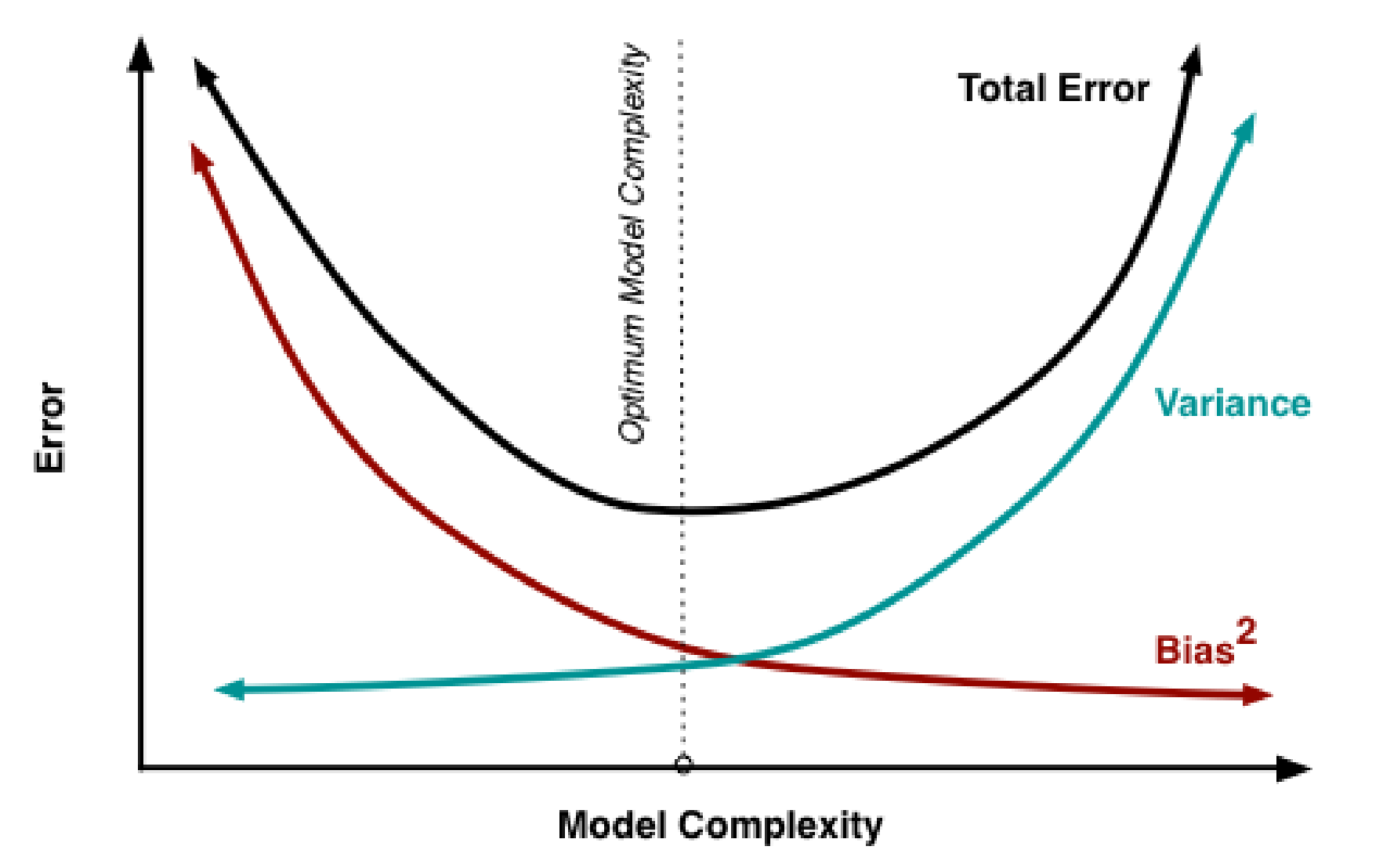
Bias and variance decomposition

- A paradigm shift: from statistical significance (R^2, t-stat, p-value) to out-of-sample forecast and bias-variance trade-off.

We assume that $Y = f(X) + \epsilon$ with epsilon such as $E(\epsilon) = 0$ and $Var(\epsilon) = \sigma^2$ then we can decompose the expected prediction error of the regression at $X = x_0$ as:

$$\begin{aligned}
Err(x_0) &= E[(Y - \hat{f}(X))^2 | X = x_0] \\
&= Var[Y|X = x_0] + E[(f(x_0) - \hat{f}(x_0)]^2 \\
&= \sigma^2 + MSE \\
&= \sigma^2 + [E(\hat{f}(x_0)) - f(x_0)]^2 + E[(\hat{f}(x_0) - E(\hat{f}(x_0)))^2] \\
&= Irriducible\ Error + Bias^2 + Variance
\end{aligned}$$

MSE is the mean squared error.


Total Error
Optimum Model Complexity
Error
Variance
Bias^2
Model Complexity

Linear Regression

Linear regression model is well known, however we would like to clarify some aspects that are important in order to understand extensions.
Given $Y \in \mathbb{R}^n$, $X \in M_{n\times k}(\mathbb{R})$ with $rank(X) = k$ and $\beta \in \mathbb{R}^k$ we define our model as:

$$Y = X\beta + \epsilon \tag{8}$$

with $E(\epsilon) = 0$ and $Cov(\epsilon) = \sigma^2 Id$
the model can be estimated using a least squares approach:

$$\hat{\beta} = \arg\min_{\beta} ||Y - X\beta||^2 \tag{9}$$

the solution is given by:

$$\hat{\beta} = (X'X)^{-1}X'y \tag{10}$$

The fitted value are then given by:

$$\hat{Y} = X(X'X)^{-1}X'y = Hy \tag{11}$$

Properties of $\hat{\beta}$

- $E(\hat{\beta}) = \beta$
- $Var(\hat{\beta}) = \sigma^2(X'X)^{-1}$

Numerically can be challenging to calculate the matrix $(X'X)^{-1}$ and the program is often solved numerically using the gradient descent algorithm.

Gradient Descent

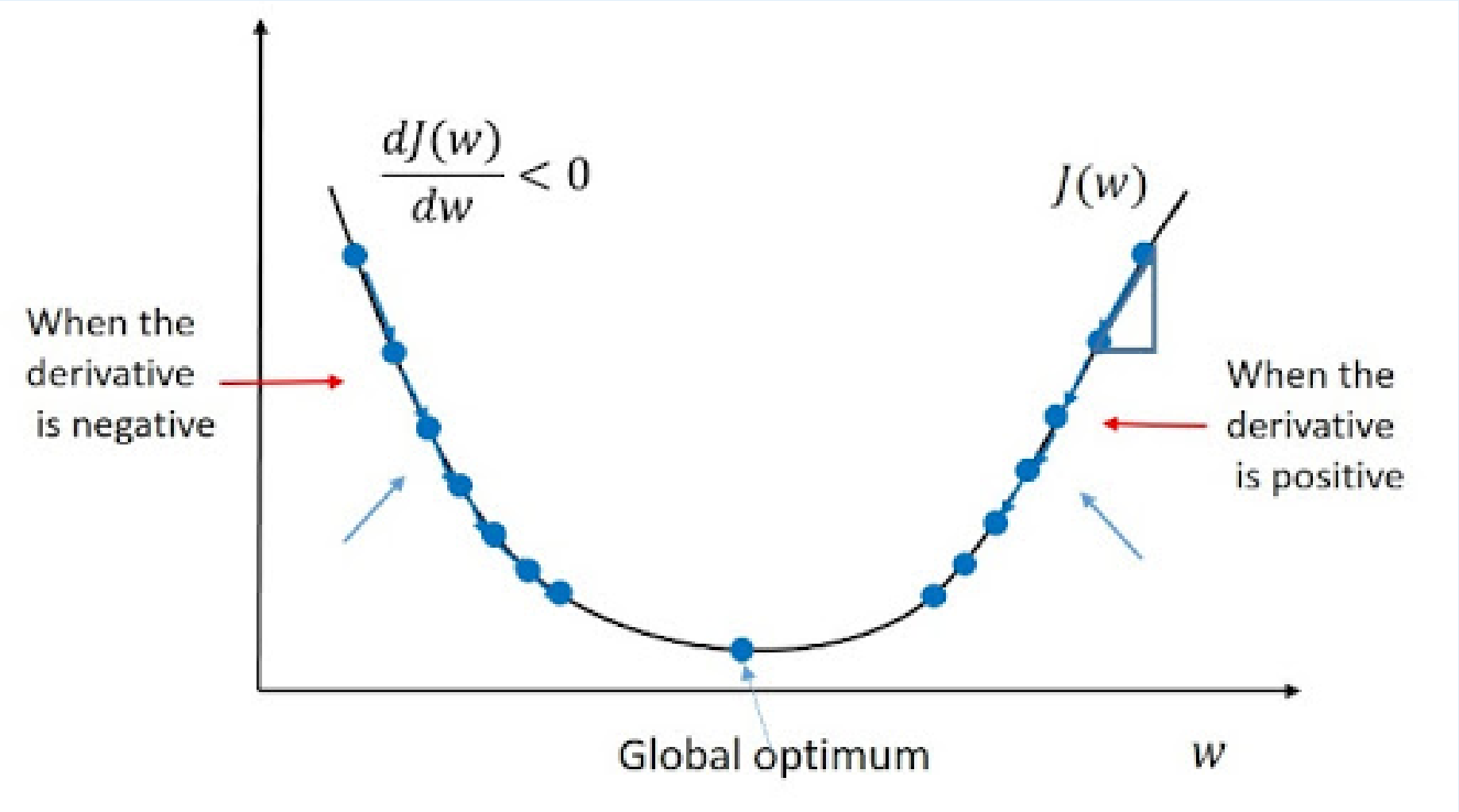
A common approach it is to use gradient descent to solve the OLS problem. In a less compact form the objective function can be written as:

$$L(\beta) = \min_{\beta} \frac{1}{n} \sum_{i=1}^{n} (y_i - \beta x_i)^2 \tag{12}$$

where we assume for simplicity that we have a univariate linear regression function without intercept. The gradient decent methodology consists in updating an initial guess value of β until convergence using:

$$\beta = \beta - \alpha \frac{\partial L(\beta)}{\partial \beta} \tag{13}$$

with α "the learning rate" fixed. In case of the OLS problem the objective function is convex hence the solution exists and it is unique.


$\frac{dJ(w)}{dw} < 0$
$J(w)$
When the derivative is negative
When the derivative is positive
Global optimum
w

Gauss-Markov theorem

The fact that $E(\hat{\beta}) = \beta$ is of particular importance since it is telling that the least square is an unbiased regressor of the true value. Let's dig a little bit more onto this.
If $\hat{\beta}$ is the estimated parameter and β^* is the true value then for any $a \in \mathbb{R}^k$, the estimate $a'\hat{\beta}$ is **unbiased** for $a'\beta^*$ if

$$E[a'\hat{\beta}] = a'\beta^* \tag{14}$$

telling us that for any future observation of X our prediction would be unbiased of the true mean. What is the best unbiased estimator?

The answer is given by the Gauss-Markov theorem. First of all we need to define a measure of best. In the Gauss-Markov we mean best in MSE (mean squared error) sense. Given the estimate $a'\hat{\beta}$

$$MSE(a'\hat{\beta}) = E[(a'\hat{\beta} - a'\beta^*)^2] \tag{15}$$

Gauss-Markov theorem If $a'\tilde{\beta}$ is another unbiased estimate of $a'\beta^*$ then

$$MSE(a'\hat{\beta}) \leqslant MSE(a'\tilde{\beta}) \tag{16}$$

where $\hat{\beta}$ is the OLS estimator.

Equivalently the Gauss-Markov theorem says that the linear regression has the smallest variance between the unbiased estimators.
Let's take a look at this in more detail. Given x_0 independent from the training set and $\hat{f}(x_0)$ a linear estimation then:

$$MSE(\hat{f}(x_0)) = Bias^2 + Var(\hat{f}(x_0)) = 0 + Var(\hat{f}(x_0)) \tag{17}$$

Question: is there a linear regressor with lower MSE?
Let's look at the average over all inputs

$$MSE(\hat{f}) = \frac{1}{n}\sum_{i=1}^{n} MSE(\hat{f}(x_i)) \tag{18}$$

For the OLS the above become

$$MSE(\hat{f}) = \frac{1}{n}\sum_{i=1}^{n} Var(\hat{f}(x_i'\hat{\beta})) = \frac{k\sigma^2}{n} \tag{19}$$

with k the numbers of explanatory variable. As we can see we are adding the same amount of variance independently from the dimension of the true coefficients.

Can we do better? The answer is yes if we relax some assumptions.

CAPM in a nutshell

CAPM is a first simple example of what we discussed in the introduction
Let's recap the univariate linear model with intercept.

$$y = \beta_0 + \beta_1 x + \epsilon \tag{20}$$

it can be shown that

$$\hat{\beta}_0 = \overline{y} - \hat{\beta}_1 \overline{x}, \quad \hat{\beta}_1 = \frac{(x - \overline{x})(y - \overline{y})}{||x - \overline{x}||^2} \tag{21}$$

$$\hat{\beta}_1 = \frac{cov(x, y)}{var(x)} \tag{22}$$

CAPM was developed by Sharpe and Lintner starting from the work on the mean-variance optimisation of Markowitz. It assume the existence of lending and borrowing at a risk-free rate of interest. In its most popular version it can be written as:

$$E[r_i] = r_f + \beta_{im}(E[r_m] - r_f) \tag{23}$$

with r_m the market portfolio.
From the above we can see that β measure the covariance between the asset and the market.

Question: What is the market?

CAPM does not completely explain the cross-section of expected asset returns.

Can you give an example?

Tutorial

- Build a simple Multi Asset Trend Following Strategy using linear regression and PCA (principal component analysis)
- We build a long-short portfolio based on next period model prediction

$$R_t = \beta_0 R_{t-3} + \beta_1 R_{t-6} + \cdots \tag{24}$$

From sklearn we will use:

- LinearRegression (sklearn.linear_model)
- StandardScaler (sklearn.preprocessing)
- PCA (sklearn.decomposition)
- Pipeline (sklearn.pipeline)

The data

We download from Yahoo finance the following series:

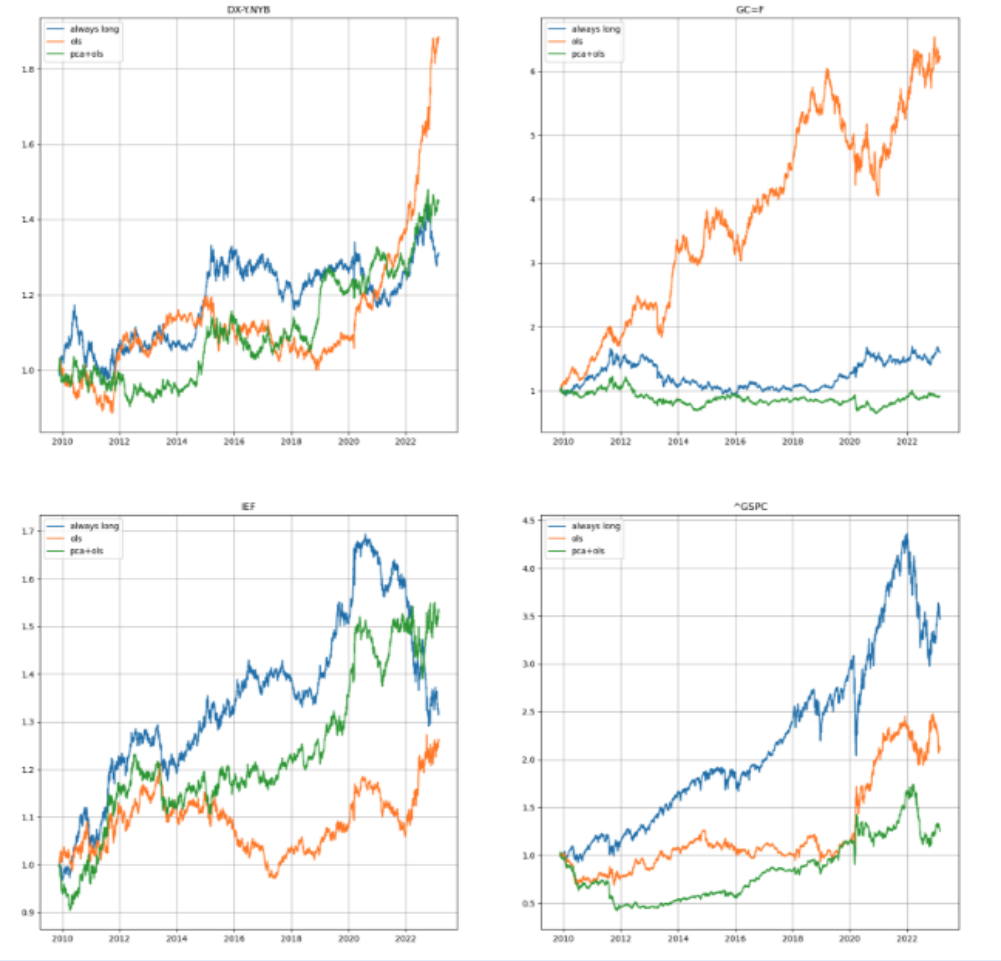
- S&P500
- iShares 7-10 Year Treasury Bond ETF
- US Dollar Index
- Gold Price

```
# Download data

# inputs
tickers = ['^GSPC', 'IEF', 'DX-Y.NYB', 'GC=F']
from datetime import datetime
start_date = '2006-11-15'
end_date = '2023-02-24'
#start_date =datetime(2006,11,15)
#end_date =datetime(2021,1,15)


# download data from yahoo finance
#panel_data = data.DataReader(tickers, 'yahoo', start_date, end_date)
panel_data = yf.download(tickers, start_date, end_date,progress=False)


# take adjusted close and compute log-returns
my_data = panel_data["Adj Close"].apply(np.log).diff()
column_names = my_data.columns
```

The exercise

- **Input features**: Previous day return Previous 1, 3, 6 and 12 months return (1 month = 21 trading days)
- **Re-estimation**: Every 63 days (3months). With sample size of 500 daily observations (rolling window)
- **Strategy** We compare buy and hold strategies with long-short strategies based on whether the next period predicted return is positive or negative

Results: Plot



Results: Sharpe ratio

$$SR = \frac{R - R_f}{\sigma} \tag{25}$$

```
# Sharpe ratio
for col in column_names:
    print('Sharpe ratio on asset ', col)
    print(' - always long is: ', np.round(np.mean(r_alwys_long[col])/ np.st
    print(' - based on ols forecast is: ', np.round(np.mean(r_ls_ols[col]),
    print(' - based on pca+ols forecast is: ', np.round(np.mean(r_ls_pca_o
```

```
Sharpe ratio on asset  DX-Y.NYB
 - always long is:  0.018
 - based on ols forecast is:  0.043
 - based on pca+ols forecast is:  0.025
Sharpe ratio on asset  GC=F
 - always long is:  0.014
 - based on ols forecast is:  0.055
 - based on pca+ols forecast is:  -0.003
Sharpe ratio on asset  IEF
 - always long is:  0.021
 - based on ols forecast is:  0.018
 - based on pca+ols forecast is:  0.033
Sharpe ratio on asset  ^GSPC
 - always long is:  0.034
 - based on ols forecast is:  0.021
 - based on pca+ols forecast is:  0.006
```

Summary

We introduced the Bias Variance decomposition and we study the OLS. In particular we showed:

- The expected prediction error can be decomposes as:

$$\begin{aligned} Err(x_0) &= E[(Y - \hat{f}(X))^2 | X = x_0] \\ &= \sigma^2 + MSE \\ &= Irriducible\ Error + Bias^2 + Variance \end{aligned}$$

- We proved the Gauss-Markov theorem. The OLS has the lowest variance among the family of linear un-biased estimator
- We showed that the OLS add the same amount of variance independently from the dimension of the coefficient

The question is. Can we reduce the expected prediction error relaxing the condition of the Gauss-Markov theorem?

Ridge Regression

Ridge regression is very similar to the ordinary least square, but shrinks the estimated coefficients of the regression $Y = X\beta + \epsilon$ Given $Y \in \mathbb{R}^n$, $X \in M_{n\times k}(\mathbb{R})$ with $rank(X) = k$ and $\beta \in \mathbb{R}^k$ we define our model as:

$$\hat{\beta}^{ridge} = \underset{\beta\in\mathbb{R}^k}{\arg\min}\, ||y - X\beta||^2 + \lambda||\beta||^2 \tag{26}$$

with $||x||^2 = \sum x_i^2$ and with $\lambda \geqslant 0$ a tuning parameter. A closed form solution exists and it is given by

$$\hat{\beta}^{ridge}(\lambda) = (X'X + \lambda 1)^{-1}X'y \tag{27}$$

The addition of a positive constant to $X'X$ before inversion makes the problem non-singular even if $Rank(X)$ is not full (i.e. k).

The distribution of the estimator $\hat{\beta}^{ridge}$ is normal and we can prove that

$$E[\hat{\beta}^{ridge}(\lambda)] = (X'X + \lambda 1)^{-1} X'X\beta \tag{28}$$

From the above we can see that the bias of the Ridge estimator is not zero in fact we have:

$$E[\hat{\beta}^{ridge}(\lambda) - \beta] = -\lambda(X'X + \lambda 1)^{-1}\beta \tag{29}$$

We can also see that:

- $lim_{\lambda \to \infty} E[\hat{\beta}^{ridge}(\lambda)] = 0$
- $lim_{\lambda \to 0} E[\hat{\beta}^{ridge}(\lambda)] = \beta$
- if X is orthogonal $(X'X = 1)$, then

$$\hat{\beta}^{ridge}(\lambda) = \frac{\hat{\beta}}{(1 + \lambda)} \tag{30}$$

hence the ridge estimator scales the OLS estimator by a factor.

In order to calculate the variance we introduce the following linear operator:

$$W_\lambda = (X'X + \lambda 1)^{-1} X'X \tag{31}$$

We can see that W_λ transforms the OLS estimator into the Ridge estimator.

$$W_\lambda \hat{\beta} = \hat{\beta}^{ridge}(\lambda) \tag{32}$$

Using the above we can calculate the variance of the Ridge estimator:

$$Var[\hat{\beta}^{ridge}(\lambda)] = \sigma^2 (X'X + \lambda 1)^{-1} X'X [(X'X + \lambda \mathbf{1})^{-1}]' \tag{33}$$

Some properties:

- $lim_{\lambda \to \infty} Var[\hat{\beta}^{ridge}(\lambda)] = 0$
- $lim_{\lambda \to 0} Var[\hat{\beta}^{ridge}(\lambda)] = \sigma^2 (X'X)^{-1}$
- if X is orthogonal $(X'X = 1)$, then

$$Var[\hat{\beta}^{ridge}(\lambda)] = \frac{\sigma^2}{(1+\lambda)^2}\mathbf{1} \tag{34}$$

We can compare the variance of the Ridge estimator with the OLS estimator and with some algebra we can show that the variance of the OLS estimator is greater or equal the variance of the Ridge estimator.

$$Var[\hat{\beta}] \geqslant Var[\hat{\beta}^{ridge}(\lambda)] \tag{35}$$

and strictly greater for $\lambda > 0$.

Bias-Variance for the Ridge regression

The above allows us to calculate the Bias-Variance decomposition that we remember is given by:

$$MSE = Trace[Var[\hat{\beta}]] + Bias^2 \tag{36}$$

For Ridge we have that:

- the Bias increases when λ becomes larger
- the Variance decreases when λ becomes larger
- The difference $MSE(\hat{\beta}^{ridge}(\lambda)) - MSE(\hat{\beta})$ can be both positive or negative. It can be proven that a value of λ for which $MSE(\hat{\beta}^{ridge}(\lambda)) \leqslant MSE(\hat{\beta})$ always exists.

Practical details

It is worth noticing that the Ridge regression is not scale-invariant. What do we mean by this?
Consider the following re-scale of our explanatory variables $Z = XR$. In the case of the OLS we have that:

$$\tilde{\beta} = (Z'Z)^{-1}Z'y = R^{-1}\hat{\beta} \tag{37}$$

i.e. the OLS solution is scale invariant. Unfortunately this is not the case for Ridge. In fact we have:

$$\tilde{\beta} = R^{-1}(X'X + \lambda(R')^{-1}R^{-1})^{-1}X'y \tag{38}$$

The general absence of scale-invariance implies that any choice we make about the scaling of variables (e.g., expressing a regressor in centimeters vs meters or thousands vs millions of dollars) affects the coefficient estimates. Since this is highly undesirable, what we usually do is to standardise all the variables in our regression, that is, we subtract from each variable its mean and we divide it by its standard deviation. When including an intercept term we leave it unpenalised. As a result the Ridge regression can be written as:

$$\hat{\beta}_0, \hat{\beta}^{ridge} = \underset{\beta_0 \in \mathbb{R},\, \beta \in \mathbb{R}^k}{\arg\min} \; ||y - \beta_0 \mathbf{1} - X\beta||^2 + \lambda ||\beta||^2 \tag{39}$$

In fact if we assume that X is standardise then $\hat{\beta}_0 = \overline{y}$.

Lasso Regression

Ridge regression can have better prediction error than OLS and it works well when some true coefficients are small, however it cannot perform variable selection. The Lasso regression is very similar, however Lasso has the capacity to set coefficients to zero. Using the same convention as before the objective function of Lasso is given by:

$$\hat{\beta}^{lasso} = \underset{\beta \in \mathbb{R}^k}{\arg\min} ||y - X\beta||^2 + \lambda ||\beta|| \tag{40}$$

with $||x|| = \sum |x_i|$ and with $\lambda \geqslant 0$ a tuning parameter. As for ridge we have that the Lasso estimator converge to the OLS for $\lambda = 0$ and $\hat{\beta}^{lasso} = 0$ for $\lambda = \infty$.

The nature of the shrinkage has two important consequences:

1. Lasso has the ability to shrink some parameters to zero: it is able to perform variable selection. Greater λ is more parameters are set to zero
2. A closed formula exists only for particular cases. A general closed formula does not exist hence we need to use numerical procedure to estimate the parameters

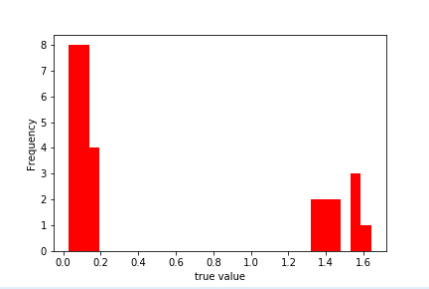
Bias and variance for the lasso estimator

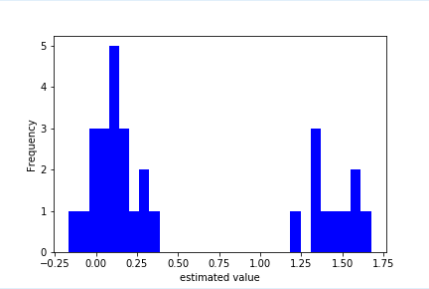
Despite a closed formula does not exist using asymptotic technique you can see that:

- The bias increases as λ increases
- The variance decreases as λ increases

As an example (taken from Tibshirani) we ran a simulation with $n = 50$ and $k = 30$. We generate $X \in \mathbb{R}^{50\times 30}$ as i.i.d. $N(0, 1)$. For a given true value of $\beta^* \in \mathbb{R}^{30}$ we generate $y = X\beta^* + \epsilon$.
We repeat the above 100 times and for any y we calculate the regression fit and we record the test error $1/n \sum_i^n (y_i - x_i'\hat{\beta})^2$.
We use the average over the 100 simulation as an estimation of the prediction error.

Bias-Variance for OLS



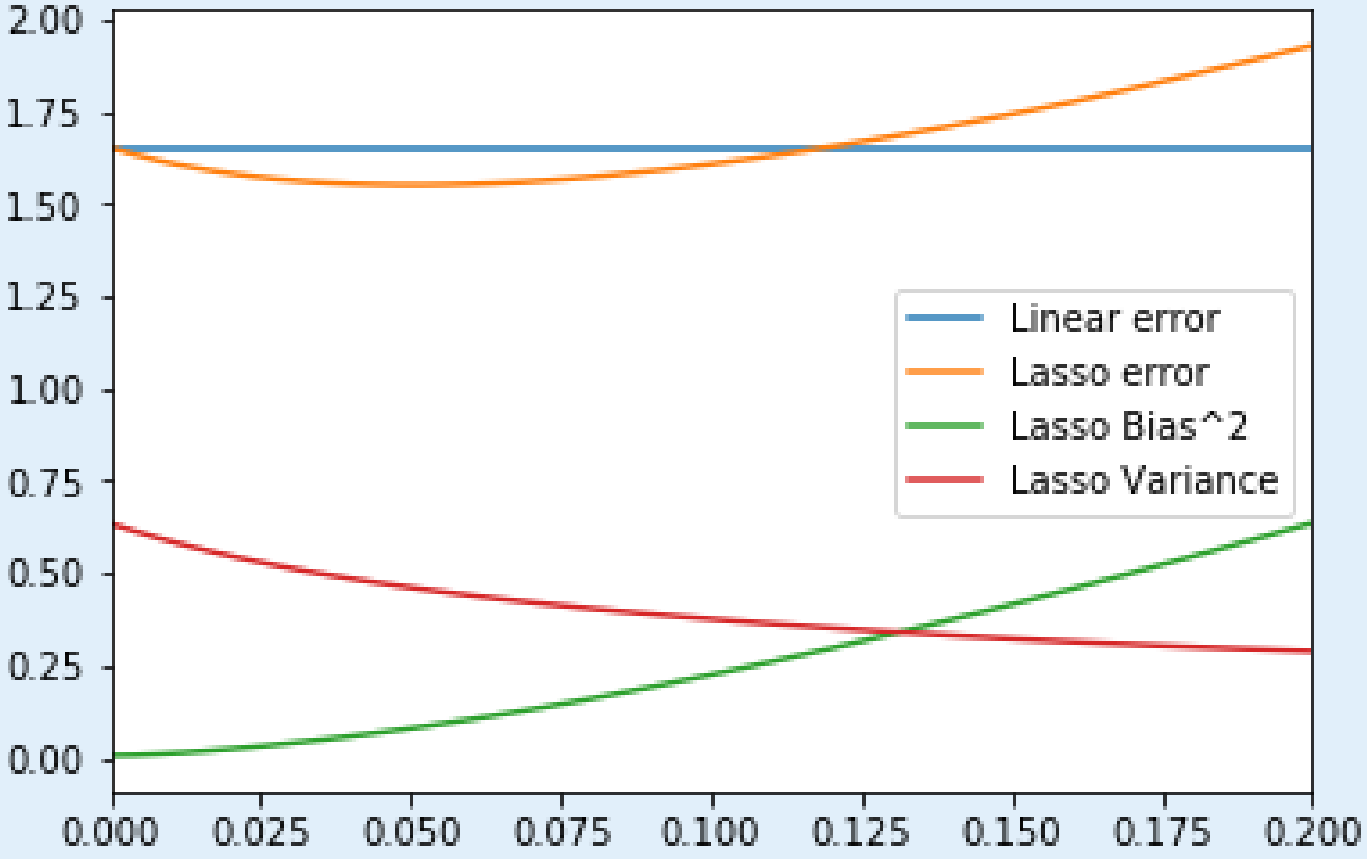


error: 1.65

noise: 1.01

bias: 0.0060

variance: 0.634

tot as sum of components: 1.65

Lasso versus linear